

Dashboard / Courses / TNE / Institut für Machine Learning / 2022W365107 / Lecture 10 + Exam (December 12, 2022)
/ Mid-Term Exam

Started on Monday, 12 December 2022, 3:32 PM

State Finished

Completed on Monday, 12 December 2022, 4:17 PM

Time taken 45 mins 12 secs

Grade 0.50 out of 10.00 (5%)

Question 1

Incorrect

Mark 0.00 out of 2.00

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t - 1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\delta(t) = \frac{\partial L}{\partial s(t)}$ is a column vector. Which of the following statements are true?

- ☐ a. The deltas fulfill the recursive relation $\delta(t) = \text{diag}(f'(s(t))) \left(V^\top \frac{\partial L(t)}{\partial \hat{y}(t)} + R^\top \delta(t+1) \right)$.
- ☒ b. The gradients for the recurrent weights R can be written as $\sum_{t=1}^T \delta(t) s^\top(t-1)$. ✗
- ☒ c. The gradients for the input weights W can be written as $\sum_{t=1}^T \delta(t) x^\top(t)$. ✓
- ☐ d. BPTT is an algorithm to compute gradients that are local in time.
- ☒ e. The memory demand for BPTT is quadratic in the sequence length T . ✗

Question 2

Partially correct

Mark 0.17 out of 1.00

Which of the following statements about RNNs are true?

- ☐ a. The number of RNN parameters grows linearly with the sequence length.
- ☒ b. Many RNN architectures are Turing complete, which means that they can learn to exactly imitate the behavior of every computer program. ✗
- ☐ c. RNNs can be applied to input sequences of varying lengths.
- ☒ d. An RNN is equivalent to a feedforward network applied to the input and an internal state representation that depends on previous time steps. ✓
- ☐ e. An RNN is equivalent to a feedforward network applied to the input at every time step separately.



Question 3

Incorrect

Mark 0.00 out of 1.00

The Elman Network (without output activation function and bias units) can be defined as

$$s(t) = Wx(t) + a(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$, and parameter matrices W, V . Which of the following statements are true?

- ☐ a. The recurrent weights are constant.
- ☐ b. The recurrent weights are equal to the identity matrix.
- ☒ c. The hidden units of the Elman network are interconnected in the sense that they can exchange information at every time step. ✖
- ☐ d. Elman proposed to train this network locally in time, i.e., on each time step separately.
- ☒ e. The recurrent Jacobian of the hidden activations is $\frac{\partial a(t)}{\partial a(t-1)} = \text{diag}(f'(s(t-1)))$. ✖

Question 4

Incorrect

Mark 0.00 out of 2.00

The LSTM (without bias units and forget gate) is defined as

$$z(t) = \tanh(Ux(t) + Ph(t-1))$$

$$i(t) = \sigma(Vx(t) + Qh(t-1))$$

$$c(t) = c(t-1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t-1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors $x(t)$, hidden activation vectors $h(t)$, memory cell state vectors $c(t)$, gate activation vectors $z(t), i(t), o(t)$, weight matrices P, Q, R, U, V, W . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. Which of the following statements are true?

- ☐ a. The gradient of the loss with respect to the hiddens is $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)} + U \frac{\partial L}{\partial z(t)} + V \frac{\partial L}{\partial i(t)} + R \frac{\partial L}{\partial o(t)}$ but it was originally proposed to be approximated by $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)}$.
- ☐ b. The memory cell fulfills $\frac{\partial c(t)}{\partial c(t-\tau)} = I$ for any $\tau \in \{1, \dots, t-1\}$, where I is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel.
- ☐ c. Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.
- ☒ d. The gradient of the loss with respect to the memory cell is $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t-1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$. ✖
- ☐ e. The LSTM architecture has no exploding or vanishing gradients because $\frac{\partial h(t)}{\partial h(t-1)}$ is an orthogonal matrix.
- ☐ f. If we choose $z(t) = \sigma(Ux(t) + Ph(t-1))$, then $E[z(t)] > 0$ and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.

Question 5

Incorrect

Mark 0.00 out of 1.00

Which statements about LSTM are true?

- ☐ a. Most LSTM networks used in practice feature a forget gate.
- ☐ b. The focused LSTM network feeds the input vector $x(t)$ only into the block input $z(t)$ and not into input gate $i(t)$ or output gate $o(t)$.
- ☐ c. To counter the drift effect, the input gate should be initialized with positive values.
- ☒ d. "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network. ✔
- ☒ e. The forget gate should be initialized with negative bias values. ✖
- ☐ f. The inputs to the lightweight LSTM have to fulfill the Markov property.
- ☐ g. An LSTM with coupled input gate and forget gate, i.e., $c(t) = (1 - i(t)) \odot c(t-1) + i(t) \odot z(t)$, leads (alongside other changes) to the gated recurrent unit (GRU).
- ☒ h. The forget gate solves the vanishing gradient problem. ✖



Question 6

Incorrect

Mark 0.00 out of 2.00

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V . Let θ denote the vector of all network parameters shared in time and let $\theta(t)$ denote their usage at time t . Further, let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial \theta}$ is a column vector.

Which statements about RTRL are true?

- ☐ a. Schmidhuber's approach divides the input sequence into chunks of size N (number hidden units) and performs RTRL within these chunks. Then it uses BPTT to consolidate the gradients for these chunks.
- ☐ b. BPTT considers the recursion $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{L(t)}{s(t)}$ while RTRL considers the recursion $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t)}{\partial s(t-1)} \frac{\partial s(t-1)}{\partial \theta}$.
- ☐ c. RTRL computes the gradients $\frac{\partial L(t)}{\partial \theta}$ during the forward pass already.
- ☒ d. The term $\frac{\partial s(t)}{\partial \theta}$ generally has $O(N^4)$ elements, where N denotes the number of hidden units. ✗

Question 7

Partially correct

Mark 0.33 out of 1.00

Backpropagation through time (BPTT) is the standard gradient computation algorithm for RNNs. However, there exist alternatives, e.g., truncated BPTT and real-time recurrent learning (RTRL). Which statements about these algorithms are true.

- ☐ a. Truncated BPTT has advantages if the input sequences are short.
- ☐ b. Truncated BPTT computes the same gradients as BPTT.
- ☒ c. RTRL is very slow and therefore rarely used in practice. ✓
- ☒ d. The runtime complexity of RTRL is $O(N^3)$ per time step, where N denotes the number of hidden units. ✗
- ☐ e. RTRL computes the same gradients as BPTT
- ☒ f. Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them. ✓

◀ LSTM10

Jump to...

LSTM11 ▶



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Started on Monday, 30 January 2023, 3:31 PM

State Finished

Completed on Monday, 30 January 2023, 4:16 PM

Time taken 45 mins

Grade 3.73 out of 10.00 (37.33%)

Question 1

Partially correct

Mark 0.93 out of 1.40

Which statements about attention mechanism are true?

- ☒ a. The scaled dot-product attention mechanism generally has the form $\text{softmax}\left(\frac{1}{\sqrt{d}}QK^\top\right)V$, where K, Q, V are keys, queries, and values, respectively, and d is the first dimension of both K and Q . ✓
- ☐ b. A spatial attention mechanism causes the network to treat each pixel in an image as equally important.
- ☒ c. An LSTM with self-attention has access to all previous hidden states and cell states via an softmax-based attention mechanism. ✓
- ☐ d. The LSTM gates can be interpreted as a temporal attention mechanism.
- ☐ e. The attention score $e(i)$ is obtained from the attention vector $a(i)$ via the softmax function.

Question 2

Partially correct

Mark 0.93 out of 1.40

Which statements about the gated recurrent unit (GRU, Cho et al., 2014) are true?

- ☐ a. The GRU has one gate more than the fully featured LSTM architecture.
- ☐ b. A forget gate mechanism is the core of the GRU architecture.
- ☒ c. In the GRU, a so-called "reset gate" $r(t)$ regulates how strongly the increments $z(t)$ depend on the hidden $h(t)$. ✓
- ☐ d. Usually, the GRU has more parameters than an LSTM with the same number of hidden units.
- ☒ e. The so-called "update gate" of the GRU acts as a coupled input and forget gate. ✓



Question 3

Incorrect

Mark 0.00 out of 1.50

Which statements about LSTM applications are true?

- ☒ a. LSTM can be used for protein classification. Hereby, the protein is represented as a sequence of amino acids and the labels are either functional or structural properties of the respective protein. ✓
- ☒ b. "word2vec" (Mikolov et al., 2013) is an example for a sentence embedding technique. ✗
- ☒ c. A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let $f(\cdot)$ be the (trained) embedding function, then $f(\text{king}) - f(\text{man}) + f(\text{woman})$ is found close to the point $f(\text{queen})$. ✓
- ☒ d. Radford et al. (2017) used an LSTM to discover sentiment in Amazon reviews by training it for sequence classification (good vs. bad) on the review text. ✗
- ☒ e. Fan et al. (2015) trained an LSTM to predict facial features from audio/text inputs. On this basis, given a speech recording, they could generate photo-realistic videos of talking heads. ✓

Question 4

Incorrect

Mark 0.00 out of 1.50

Which of the following statements about LSTM applications are true?

- ☒ a. Karpathy and Fei-Fei (2015) propose to perform image captioning using a CNN image encoder network in combination with an LSTM next-character text generation network. ✗
- ☒ b. Alex Graves showed in his 2013 paper how LSTM can be used to generate text sequences simply by predicting one character at a time. ✓
- ☒ c. Sequence-to-sequence learning tasks without one-to-one correspondences can be tackled using an encoder-decoder architecture. ✓
- ☒ d. Language translation is an example for sequence-to-sequence learning tasks with one-to-one correspondences. ✗
- ☐ e. Part-of-speech tagging is an example for sequence-to-sequence learning with one-to-one correspondences.
- ☒ f. Hochreiter et al. (2001) showed how RNNs can "learn to learn" by arranging labelled datasets as sequences. At time t , the RNN takes an input x_t along with the label y_t and makes a prediction $\hat{y}_t$. ✗

Question 5

Partially correct

Mark 0.93 out of 1.40

Which statements about Transformer networks (Vaswani et al., 2017) are true?

- ☒ a. Without positional embeddings, the Transformer network has no way of figuring out the order of elements in an input sequence. ✓
- ☐ b. The Transformer architecture consists of an encoder and a decoder part, where the encoder depends on the output of the decoder.
- ☒ c. Transformer networks use multi-head attention, which works as follows. We run multiple instances of scaled dot-product attention, concatenate their outputs (or context) and feed the result into a linear layer. ✓
- ☐ d. The Transformer architecture suffers from the vanishing gradient problem in the time dimension.
- ☒ e. BERT (Devlin et al., 2019) uses two different pre-training objectives, (1) inserting missing (masked out) words into a sentence, and (2) next sentence prediction, i.e., generating a response sentence to a question sentence. ✗



Question 6

Incorrect

Mark 0.00 out of 1.40

Which statements about LSTM variants are true?

- ☒ a. Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state. ✓
- ☒ b. A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction. ✓
- ☒ c. The multidimensional LSTM (Graves et al., 2007) can process sequences consisting of vectors which stem from high-dimensional vector spaces. ✗
- ☒ d. Stacked LSTM (Graves et al., 2013) stack LSTM layer by using one LSTM's final cell state as the initial cell state for the next LSTM. ✗
- ☒ e. Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos. ✓

Question 7

Partially correct

Mark 0.93 out of 1.40

Which statements about LSTM dropout are correct?

- ☐ a. LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients.
- ☐ b. It is especially important to apply dropout when evaluating the model.
- ☐ c. LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability.
- ☒ d. Dropout is a regularization technique that randomly sets activations to zero during the forward pass. ✓
- ☒ e. The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content. ✓

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Jump to...

[Lecture 1, 05.10.2020 ▶](#)